SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

20/20

Industry Problem Solving Task

Code of Conduct:

I K. Yamini (192572181) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
K. Y

Annexure A

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed

at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the

industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination	Good collaboration	Limited participation	No collaboration or contribution

CO4-AT-2

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1. Star Topology for a 20-Person Startup

Problem Analysis

A startup with around 20 employees needs a network that is easy to install, manage, troubleshoot, and expand. The network should connect computers, printers, VoIP phones, laptops, and other devices while providing reliable Internet access.

Proposed Solution

A star topology is the most suitable solution. In this topology, all end devices are connected to a central network switch.

Network Components

- 24-port managed Layer 2/Layer 3 switch
- Router/firewall
- Wi-Fi access point
- Computers and laptops
- Network printers
- VoIP phones
- Cat6 UTP cables
- Internet connection

Working

Each workstation is connected directly to the central switch using a Cat6 cable. The switch connects to the router/firewall, which provides Internet connectivity, DHCP, and NAT. Wireless devices connect through an access point connected to the switch.

Advantages

1. Easy to install and configure.
2. Easy to identify network failures.
3. Failure of one cable affects only one device.
4. Centralized network management is possible.

5. Additional computers can easily be added.

6. Cat6 supports high-speed Gigabit communication.

7. The network can be expanded as the startup grows.

Limitation

The central switch is a critical component. If the switch fails, communication between connected devices can be interrupted.

Conclusion

Star topology is an appropriate choice for a 20-person startup because it provides simple management, fault isolation, reliability, and scalability.

2. Mesh Topology for a Banking Network

Problem Analysis

Banks handle critical financial transactions and customer information. Therefore, the network must provide high availability, reliability, security, and minimum downtime.

Proposed Solution

A partial mesh topology is suitable for the bank's core network. A full mesh can be used where maximum redundancy is required.

Network Components

- Core routers
- Core Layer 3 switches
- Enterprise firewalls
- Servers
- Fiber-optic cables
- UPS systems
- RAID storage
- Redundant Internet connections

Working

Core routers, switches, and firewalls are connected using multiple paths. If one connection fails, network traffic can use another available path.

Protocols such as OSPF or BGP can be used for routing and automatic path selection.

Transmission Media

Fiber-optic cables are preferred because they:

- Provide high bandwidth.
- Support high-speed communication.
- Have low latency.

- Are resistant to electromagnetic interference.
- Are suitable for long-distance connections.

Single-mode fiber can be used between buildings, while multimode fiber can be used within a data center.

Advantages

1. Very high reliability.
2. Multiple alternative communication paths.
3. Failure of one link does not necessarily stop communication.
4. Supports automatic traffic rerouting.
5. Suitable for critical financial transactions.
6. Provides high network availability.
7. Supports future network expansion.

Limitation

Mesh networks are expensive because they require more cables, ports, network devices, and configuration.

Conclusion

Mesh topology is suitable for banking because redundancy and fault tolerance are more important than low cost.

3. Bus Topology for a Small Classroom Lab

Problem Analysis

A small classroom laboratory may have a limited budget. The network should provide basic connectivity for computers, file sharing, and Internet browsing without requiring expensive networking equipment.

Proposed Solution

A bus topology can be considered because all computers share a common communication path.

Network Components

- Common backbone cable
- RG-58 coaxial cable
- T-connectors
- BNC connectors
- 50-ohm terminators
- Computers
- Network interface cards

Working

In a traditional bus network, all computers are connected to one main communication cable. Data transmitted by one computer travels along the shared bus, and the intended device receives the data.

Terminators are installed at both ends of the cable to prevent signal reflection.

Advantages

1. Low hardware cost.
2. Requires less cabling.
3. Simple design.
4. Suitable for small networks.

5. Does not require a central switch in the traditional implementation

6. Easy to understand and implement in a basic lab

Disadvantages

1. Cable failure can affect the entire network.

2. Network performance decreases as more devices are added.

3. Collisions can occur.

4. Troubleshooting can be difficult.

5. Limited scalability.

6. Not suitable for large organizations.

Practical Use

For a classroom with light network usage, the low cost can justify these limitations. The uploaded task also describes a contemporary low-cost alternative using an inexpensive switch and Cat5e patch cables, while noting that this is technically more switch-based than a classic bus.

Conclusion

Bus topology can be suitable for a small, low-budget, supervised classroom environment, where network traffic is relatively low.

4. Hybrid Topology for a University Campus

Problem Analysis

A university campus contains thousands of users distributed across multiple buildings such as:

- Lecture halls
- Computer laboratories
- Libraries
- Hostels
- Administration buildings
- Research centers

Using only one topology for the entire campus would not be efficient.

Proposed Solution

A hybrid topology combines different network topologies according to the requirements of each area.

Network Structure

Campus Core → Building Distribution → Floor Access → End Devices

Campus Backbone

A fiber-based ring or partial mesh can be used at the campus backbone.

High-speed fiber connects core routers and multilayer switches.

Building Network

Each building uses a star topology. A distribution switch connects to floor-level access switches.

Floor Network

Access switches connect computers, printers, and other devices using Cat6 cables.

Wireless Network

Wi-Fi access points are connected to access switches using Power over Ethernet (PoE).

Advantages

1. High scalability
2. Easy to add new buildings.
3. Provides backbone redundancy.
4. Easy management within buildings.
5. Supports wired and wireless devices.
6. Reduces the impact of individual link failures.
7. Provides high-speed communication through fiber.
8. Suitable for thousands of users.
9. Supports future expansion.
10. Combines the advantages of different topologies.

Example

If a university builds a new engineering block, a new distribution switch can be installed and connected to the campus core using fiber without redesigning the entire existing network.

Conclusion

Hybrid topology is the best choice for a large university because it provides redundancy at the backbone, simplicity within buildings, and scalability across the campus.

5. Network Design for a Company with Multiple Departments

Problem Analysis

A company may have departments such as:

- Human Resources
- Finance
- Sales
- Information Technology

Each department has different users, computers, printers, and security requirements. The network must provide communication while preventing unnecessary access between departments.

Proposed Solution

A hierarchical network architecture using VLANs can be implemented.

Network Layers

Access Layer → Distribution/Core Layer → Router/Firewall → Internet

Access Layer

Each department has a dedicated Layer 2 switch.

For example:

- HR → VLAN 10
- Finance → VLAN 20
- Sales → VLAN 30
- IT → VLAN 40

VLAN numbers are example values and can be changed according to the organization's IP plan.

Core Layer

A Layer 3 switch or router performs inter-VLAN routing. This allows authorized communication between different departments.

Internet Connectivity

A router connects the internal company network to external networks and the Internet. Firewall functionality can be used to protect the internal network.

Security

VLANs provide logical separation between departments. Access-control rules can be applied to restrict sensitive resources such as Finance servers.

IP Addressing

Each VLAN can have its own IP subnet. This makes network administration easier and reduces unnecessary broadcast traffic.

Reliability

Redundant links and backup paths can be added to reduce downtime if a network link fails.

Advantages

1. Better network security.
2. Reduced broadcast traffic.
3. Easy network management.
4. Department-wise logical separation.
5. Efficient communication.
6. Easy troubleshooting.
7. Supports redundancy.
8. Scalable for future growth.
9. Better control over network resources.
10. Suitable for medium and large organizations.

Conclusion

A hierarchical VLAN-based network provides security, performance, reliability, manageability, and scalability. It allows different departments to operate independently while still communicating when required.